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The effect of increasing dietary palmitic and stearic acid on melting properties of milk fat from Holstein cows

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ABSTRACT

Palmitic and stearic acid are commonly fed to dairy cows but limited data are available on their effects on thermal properties of milk fat, especially when fed at different levels. Recently, some consumers voiced concerns about butter being harder at room temperature and questioned if it was caused by palmitic acid supplementation of dairy cows. Our hypothesis was that increasing palmitic acid intake would linearly increase palmitic acid in milk fat and, therefore, increase the solid fat content (SFC) of milk fat at 20°C, whereas increasing stearic acid intake would increase both stearic and oleic acid in milk fat and not change the SFC of milk fat at 20°C. A total of 12 second-lactation Holstein cows (106 ± 31 DIM) were arranged in a replicated 3 × 3 Latin square design with a dose escalation design within period and ≥10 d washouts between periods. Treatments included a no-supplement fat control (CON), a fatty acid (FA) supplement high in palmitic acid (PA; 88% palmitic and 8% oleic), and a FA supplement high in stearic acid (SA; 81% stearic and 10% oleic). The FA supplements were fed at increasing doses every 4 d, targeting 150, 300, 500, and 750 g/d. Milk samples were collected on d 3 and 4 of each dose, composited, and milk fat extracted from the fat cake by centrifugation. The FA profile of whole milk was analyzed using GC, and the melting properties of milk fat were characterized by differential scanning calorimetry. Data were analyzed by ANOVA with preplanned contrasts testing CON versus PA and CON versus SA at each dose level. Increasing PA progressively increased palmitic acid in milk fat, whereas increasing SA progressively increased both stearic and oleic acid. At 750 g/d, PA increased palmitic and palmitoleic acid in milk fat 5.7 and 0.25 percentage units compared with CON, whereas SA increased stearic and oleic acid in milk fat by 2.4 units and 3.0 units compared with CON. The SFC of milk fat at 20°C was linearly increased by PA but was decreased by SA. At the 750 g/d dose, PA increased SFC by 5.3 percentage units, whereas SA decreased it 3.2 percentage units compared with CON. In conclusion, increasing palmitic acid intake increases the SFC of milk fat at room temperature, whereas increasing stearic acid modestly decreases it, likely due to differences in the rates of desaturation of these FA by the stearoyl-CoA desaturase enzyme.

Key words: butter, differential scanning calorimetry, melting properties

INTRODUCTION

Milk is largely a commodity, and although the price paid to most dairy producers in the United States and Canada is often influenced by milk fat concentration, it is not affected by fatty acid (**FA**) profile, manufacturing properties, or other characteristics that may be of value to consumers. Thus, producers have a strong interest in increasing milk fat concentration and yield, and fat supplementation is a common strategy used to do so (Loften et al., 2014). Saturated FA supplements are commonly fed and allow increasing dietary fat without increasing the risk of biohydrogenation-induced milk fat depression that is associated with dietary UFA. Commercially available saturated FA supplements differ in their enrichment of palmitic (16:0) and stearic (18:0) acid, and palmitic acid supplements have become popular because they more consistently increase milk fat yield (Loften et al., 2014). These supplements have well-documented effects on milk FA profile (dos Santos Neto et al., 2021), but they can also influence product qualities important to customers, such as butter melting properties.

Butter is a heterogeneous mixture of highly diverse triglycerides that differ in melting temperature, and it thus has a broad and complex melting curve that is especially dynamic around room temperature. This influences spreadability and plasticity and may cause butter to be too soft or too firm, or to melt or soften too quickly or too slowly, depending on the application. The melting

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

properties of milk fat have been extensively investigated, but the focus has been on processing methods to modify functionality (Kaylegian and Lindsay, 1995), with little work on differences due to preharvest factors such as diet composition. The melting temperature of milk fat is influenced both by FA profile, as FA differ greatly in melting temperature, and the position of FA within each triglyceride. Briefly, increasing chain length of SFA increases the melting temperature, and increasing the number of double bonds decreases the melting temperature. Because over half of the FA in milk fat originate directly from absorption of dietary fat (Harvatine et al., 2009), there is a large and direct link between absorbed FA and milk FA profile.

Despite known changes to milk FA profile with fat supplementation, little research has analyzed the direct effect of dietary fat supplementation on the melting properties of milk fat. For consumers, texture (and more specifically, spreadability) is one of the most important functional properties of butter (Krause et al., 2007) and is largely influenced by its melting behavior. Although further research is needed to more clearly understand consumer preferences for specific attributes of butter, US butter imports, especially from Ireland, have grown tremendously (Simoes and Hidalgo, 2011). This growth is largely due to strong demands for premium butter (Liebrand, 2022) and may indicate a consumer preference for butter that differs in melting temperature, color, or flavor from domestic stocks. Consumer dissatisfaction with North American butter was also expressed in a Twitter post by a cookbook author in 2019 that asked if anyone had noticed butter was "no longer soft at room temperature," as well as being "watery" and "rubbery" (Van Rosendaal, 2021). The post went viral and sparked international media coverage. The subsequent discussion proposed that increased feeding of palmitic acid supplements to dairy cows as a possible cause. Although a limited number of studies have indicated that palmitic acid supplementation increases melting temperature (summarized by Harvatine (2021)), these experiments were generally conducted at a higher feeding rate, and information on potential differences in the magnitude of these responses with varying degrees of palmitic acid supplementation is lacking. This understanding would be very useful, as decisions on how much palmitic acid to include in the diet often vary based on region, the market price of supplements, and the premiums received for milk fat. Generally, fat supplement feeding rates range from zero to 750 g/d, with farms most frequently feeding 100 to 500 g/d. There is also a need to compare palmitic acid supplementation to stearic acid, another common SFA supplemented in dairy cow rations. The objective of the current study was to determine the dose-dependent effect of increasing dietary palmitic and stearic acid on the milk FA profile and melting properties of milk fat. Supplementing FA at increasing doses provides greater variation of milk FA profile than is typically observed at one feeding rate in order to better characterize the relationship between milk FA profile and milk fat melting properties. Our hypothesis was that increasing palmitic acid intake would linearly increase palmitic acid in milk fat and increase the solid fat content (**SFC**) of milk fat at room temperature, whereas increasing stearic acid intake would increase both stearic and oleic acid in milk fat and not change the SFC of milk fat room temperature due to desaturation of a portion of stearic acid to oleic acid in the mammary gland.

MATERIALS AND METHODS

Experimental Design

A total of 12 second-lactation Holstein cows (106 ± 31 DIM) were arranged in a replicated 3 × 3 Latin square with a dose escalation design within each period and a ≥10 d washout between periods. Treatments included a no-supplement control (**CON**), an FA supplement high in palmitic acid (**PA**; 88% palmitic acid, 1% stearic acid, 8% oleic acid, and 3% other FA as a mixture of Ca salt and free FA prills; Spectrum Fusion, Perdue Agribusiness), and an FA supplement high in stearic acid (**SA**; 81% stearic acid, 10% oleic acid, 7% palmitic acid, and 2% other FA; custom free FA prill, Actus Nutrition, Eden Prairie MN). The FA supplements were fed at increasing doses every 4 d, targeting 0, 150, 300, 500, and 750 g/d of supplemental FA as explained in the following section (Figure 1). The washout between periods 2 and 3 was longer than periods 1 and 2 (19 and 10 d, respectively) to allow more time to transition to a new silage source.

Supplement Feeding

Treatments were fed once daily as a TMR targeting a 5% refusal rate. On the first day of each dose period, the average feed intake over the previous 4 d was determined for each treatment. The TMR concentration of each FA supplement was calculated to provide the targeted dose and remained constant across all 4 d of each dose level unless animal intakes changed by more than ~10%, in which case the average intake per treatment and amount of supplement was recalculated. The PA supplement contained some calcium salts of FA and thus had a small amount of calcium (~1%). Limestone was added to the SA treatment to provide an equal amount of calcium.

Milk Sampling and Processing

Milk samples were taken at both milkings on d 3 and 4 of each dose, composited according to milk yield, and mixed by inverting at least 5 times. An aliquot was

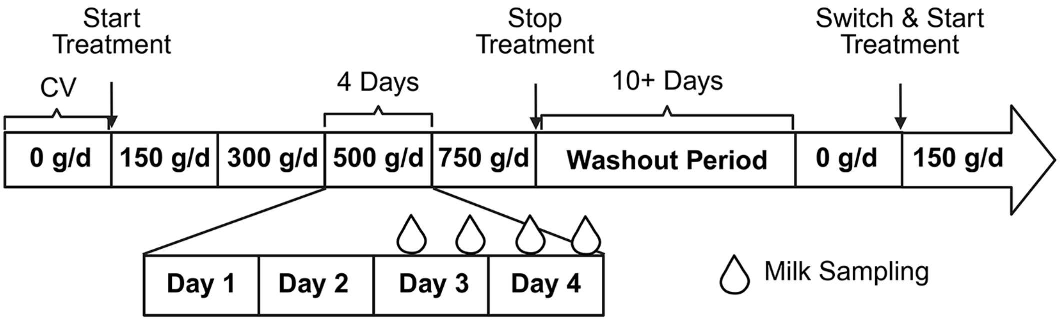

Figure 1. Experimental design; 12 second-lactation Holstein cows (106 ± 31 DIM) were arranged in a replicated 3 × 3 Latin square with a dose escalation design within each period and a ≥10 d washout between periods. Treatments were a no-supplement control (CON) or fat supplements high in palmitic acid (PA; 88% 16:0 and 8% 18:1*c*9) or stearic acid (SA; 81% 18:0 and 10% 18:1*c*9) fed at increasing doses every 4 d (0, 150, 300, 500, or 750 g/d). Milk samples were collected during both PM and AM milkings for each dose level within each period and composited across d 3 and 4. Samples taken during the 0 g/d dose each period were used as a covariate for milk FA analysis. Createdwith BioRender.com. Homan, A. (2025); https://biorender.com/t1usz7i.

immediately pipetted into a glass extraction tube for FA extraction and analysis of milk FA profile, and the remainder was centrifuged at 1,300 × *g* at 4°C for 15 min. The resulting fat cake was harvested and stored in screw-top cryogenic vials at −20°C.

Milk Fatty Acid Analysis

Milk FA were extracted using hexane:isopropanol, transmethylated with sodium methoxide, and quantified using a GC with a flame ionization detector as described by Baldin et al. (2018). Individual FA correction factors were used to account for average GC recovery, as well as the proportional weight of the methyl ester group on each FA, as described by Rico and Harvatine (2013).

Preparation for Milk Fat Melting Analysis

Milk fat was extracted from the fat cake by centrifugation. The resulting sample is similar to butteroil or anhydrous milk fat. Briefly, the fat cake was heated at 60°C in a forced air oven for ~20 min until it melted and was then centrifuged at 21,000 × *g* for 20 min at room temperature. The top layer containing melted milk fat was transferred into a glass vial using a glass Pasteur pipette that was prewarmed in a 40°C oven. Samples were then cooled to room temperature and stored at −20°C. Fat cake samples that did not have clear separation between milk fat and other solids were stored at 4°C overnight, reheated to 65°C, and centrifuged againbefore transferring as described previously. Approximately 12% of samples had unusual distortions in the differential scanning calorimetry (**DSC**) curve (method in the following section) that included generally large, uncharacteristic spikes, and white particles were visually observed in the milk fat, likely due to contamination of the fat layer when transferring layers. These samples were re-extracted by heating to 70°C, recentrifuged, and processed as described previously. The re-extraction resulted in normal DSC curves that were used in the final analysis in place of the original.

Analysis of Milk Fat Melting Properties

Melting properties of the milk fat were analyzed using DSC (Q600 SDT, TA Instruments). Samples were heated for ~20 min at 60°C in a heat block to melt the milk fat. The milk fat was then briefly vortexed, and 2 to 3 mg were pipetted into aluminum Tzero pans and lids (TA Instruments) using a warmed 10-μL plastic pipette tip. An empty aluminum pan was used as a reference. The DSC melting test was similar to that described by Tomaszewska-Gras (2013). Samples were heated to 50°C for 5 min and cooled at 5°C/min to −40°C and then held for 5 min for controlled crystallization. Samples were then heated at 5°C/min to 60°C and data recorded.

Melting characteristics were quantified using TRIOS Softwarefrom TA Instruments (example shown in Figure 2). The "Peak Integration (Enthalpy)" function was used to draw a baseline from the plateau of the melting curve when the sample was fully melted to the start of melting. The area under this curve was calculated to quantify the enthalpy of melting (Shepardson et al., 2020), and peak melting temperature was designated as the nadir of the melting curve. The "Running Integral" function was then used to record enthalpy at 5°C increments from 0°C to 40°C, as well as 3°C to represent refrigerator temperature and 38.6°C to represent cow body temperature. The SFC at each temperature was calculated at each point as (total enthalpy − enthalpy up to X°C)/total enthalpy. Finally, the "Endset Point" function was used to determine the end melting temperature, defined as the inflection just before final melt at the end of the melting curve.

Statistical Analysis

All statistical analyses were conducted in JMP Pro 15 (SAS Institute). First, ANOVA was conducted with a model that included the random effects of cow and period (1 to 3) and the fixed effects of dose (150, 300, 500, and 750 g/d), treatment (CON, PA, and SA), and their interactions. Dose was categorized as an ordinal variable. For milk FA data, the 0 g treatment for each period was used as a covariate in the model. The 0-g samples were not analyzed for melting properties and therefore were not available as a covariate. Preplanned contrasts tested CON versus PA and CON versus SA within each dose level. Significance was declared at $P \leq 0.05$ and tendencies at $0.05 < P \leq 0.10$.

Second, changes in the solid fat curves (repeated observations of SFC across temperature) were analyzed by ANOVA with a model that included the random effects of cow and period and the fixed effects of treatment, dose, temperature, and their 2- and 3-way interactions. The model failed to fit with either raw or log-transformed data, and it was determined that only 35°C to 40°C required log-transformation. Therefore, the dataset was split, with 0°C to 30°C was analyzed together, and 35°C to 40°C log-transformed and analyzed together. The log-transformed data were back-transformed for presentation. The significance of the main effects and interactions from the 0°C to 30°C model are reported, as this is the portion of the curve of applied interest.

In all analyses, data points outside of ±3.25 Studentized residual were considered outliers and removed. Two samples were outliers for SFC of milk fat at room temperature. Because this is the main variable of interest, these samples were removed from all variables, as well as milk FA analysis. Additionally, one sample was lost during milk fat extraction and one cow only had samples collected during 2 periods (PA and SA treatment) after replacing another cow after period 1 that had poor temperament. The final dataset included 137 samples.

Regression analysis was conducted to further investigate relationships between individual FA and SFC of milk fat at room temperature. First, a bivariate analysis tested the linear and quadratic relationship between each individual FA and SFC of milk fat at room temperature. The quadratic term did not considerably improve model fit for any FA (<0.1 change in adjusted R^2), so only linear relationships were analyzed further. Individual FA that had concentrations ≥1% of total FA and an adjusted R^2 >0.40 were analyzed using random regression to determine the proportion of variance due to cow and period, as FA <1% of milk FA likely have a limited direct effect on milk fat melting due to low concentrations. The 2 FA that fell into this category were 16:0 and 18:1*c*9. Partial R^2 for 16:0 and 18:1*c*9 were calculated in random regression models by subtracting the R^2 of the reduced model containing just random effects from the full model for each. Data points were adjusted for the effect of cow and period using the Conditional Predicted formula in JMP to make adjusted SFC plots.

Backward multivariate regression was conducted between SFC of milk fat at 20°C and milk FA present at >1% of FA (4:0, 6:0, 8:0, 10:0, 12:0, 14:0, 14:1*c*9, 15:0, 16:0, 16:1*c*9, 18:0, 18:1*c*9, 18:1*c*11, and 18:2*c*9,*c*12). The model was reduced by removing the least significant FA until all model factors were $P \leq 0.01$, Backward regression analysis was also conducted with the FA as a molar percentage of FA.

Finally, principal component analysis (**PCA**) was conducted to explore relationships between the concentrations of the individual milk FA and SFC of milk fat at 20°C. Two outliers were removed from the dataset for PCA due to having a T^2 value >30.

RESULTS

Profile of FA Supplements

Supplements were expected to contain ~80% palmitic or stearic acid and ~10% oleic acid. However, the PA supplement contained 87.7% palmitic acid, 1.19% stearic acid, 7.62% oleic acid, and 3.46% other FA. The SA supplement contained 6.52% palmitic acid, 81.1% stearic acid, 9.80% oleic acid, and 2.58% other FA.

Intake of FA Supplements

Treatments were fed targeting an average intake of 150, 300, 500, and 750 g/d of FA for each supplement. The amount of FA in each supplement consumed based on actual intake and supplement FA percent was 145 ± 12.7, 293 ± 25.1, 484 ± 34.8, and 722 ± 57.1 g/d for the PA treatment and 150 ± 11.3, 307 ± 26.7, 503 ± 34.0, and 757 ± 59.9 g/d for the SA treatment (mean ± SD).

Milk FA by Category

Palmitic acid reduced the concentration of FA <16C in milk fat starting at 150 g/d compared with control, whereas SA reduced them starting at 300 g/d (Table 1). The magnitude of the decrease was greater for PA than SA at all dose

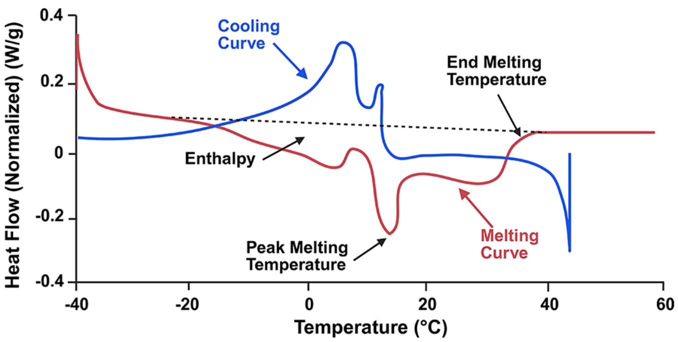

Figure 2. Example of enthalpy curves of milk fat using differential scanning calorimetry (DSC). Samples were heated to 50°C for 5 min, cooled at 5°C/min to −40°C (blue cooling curve) and then held for 5 min for controlled crystallization. Samples were then heated at 5°C/min to 60°C (red melting curve). Data were recorded for the melting curve, which included peak melting temperature, end melting temperature (using the endset function), and enthalpy (represented by the area between the black dotted line and the red melting curve). Created with BioRender.com. Homan, A. (2025); https://biorender.com/tadl8te.

levels, with a maximum decrease of 3.80 percentage units for PA ($P < 0.001$) and 1.38 percentage units for SA ($P = 0.003$). The concentration of 16C FA was increased up to 7.3 percentage units by PA starting at 150 g/d, whereas SA decreased 16C FA up to 3.32 percentage units starting at 300 g/d ($P < 0.001$ for both; Table 1). The concentration of FA >16C was increased by SA up to 5.4 percentage units starting at 300 g/d, whereas PA decreased >16C FA up to 2.4 percentage units starting at 500 g/d (Table 1). Both PA and SA decreased the concentration of total odd- and branched-chain FA (**OBCFA**) starting at 150 g/d (up to 0.88 and 0.66 percentage units, respectively; $P < 0.05$ for both; Table 1), with similar patterns for both odd-chain FA and branched-chain FA (data not shown).

Two of the biggest determinants of melting temperatures for butter are total saturated versus unsaturated fat concentration and mean chain length. The total concentration of SFA was increased up to 0.89 percentage units by PA starting at 500 g/d, whereas SA decreased total saturated FA up to 2.28 percentage units, also starting at 500 g/d (Table 1). Conversely, SA increased UFA concentration up to 1.08 percentage units and PA decreased UFA 2.17 percentage units, both starting at 500 g/d. The mean carbon chain length was calculated based on the percentage of each known FA on a molar basis and averaged 13.48 C across CON samples. The PA treatment had a higher mean carbon chain length than CON at 300 and 750 g/d, whereas SA had a greater mean carbon chain length than CON at 300 g/d and above (Table 1). At 750 g/d, this difference was 0.15 for PA ($P = 0.002$) and 0.12 for SA ($P = 0.01$).

Selected Milk FA of Interest

Within the <16C FA, there was no effect of PA or SA on 4:0 (Table 2). However, PA decreased the concentration of 6:0 at 300 and 750 g/d and decreased the concentration of 8:0, 10:0, 12:0, and 14:0 at all dose levels compared with CON. We found no effect of SA on 6:0 or 8:0, but it reduced the concentration of 10:0, 12:0, and 14:0 starting at 300 g/d. Finally, both PA and SA reduced the concentration of 14:1*c*9 starting at 500 g/d ($P < 0.05$).

Palmitic acid concentration in milk FA was increased up to 7.0 percentage units by PA and decreased up to 3.0 percentage units by SA. Similarly, PA increased the concentration of palmitoleic acid (16:1*c*9) in milk fat at all dose levels (up to 0.31 percentage units), whereas SA decreased it starting at 300 g/d (up to 0.29 percentage units; Table 3).

The concentration of stearic acid in milk FA was increased up to 2.6 percentage points starting at 300 g/d of SA ($P < 0.001$), whereas PA only decreased stearic acid concentration at the 750 g/d dose (1.2 unit decrease, $P < 0.001$). The SA supplement also increased oleic acid up to 2.8 percentage points starting at 300 g/d of supplementation ($P < 0.001$). Oleic acid tended to be decreased by PA at 500 and 750 g/d ($P < 0.10$; Table 3).

We found no effect of SA on the concentration of 18:2*c*9,*c*12 or 18:3*c*9,*c*12,*c*15 in milk fat, but PA decreased both starting at 500 g/d. The decrease in 18:2*c*9,*c*12 was 0.15 percentage units ($P = 0.004$) and 18:3*c*9,*c*12,*c*15 was 0.09 units ($P = 0.002$) at the 750 g/d dose. We found no effect of PA or SA on CLA*c*9*t*11 (Table 3).

Desaturation and Butter Spreadability Indices

The desaturase index is calculated as the product to precursor ratio of the stearoyl Co-A desaturase enzyme (Kelsey et al., 2003). The desaturase index for 16-C FA, calculated as [palmitoleic %/(palmitoleic % + palmitic %)] × 100, averaged 4.54 for the CON. We found no effect of PA on the 16-C FA index, but SA decreased it up to 0.48 units starting at 300 g/d. The 18-C desaturase index, calculated as oleic %/(oleic % + stearic %) × 100, averaged 65.1 across observations for CON and was increased up to 2.6 units by PA ($P = 0.01$), whereas SA tended to decrease it by 1.7 units compared with CON ($P = 0.08$; Table 4).

Butter spreadability index (**SI**) is defined as the ratio of palmitic acid to oleic acid (16:0/18:1), with higher values indicating harder butter with reduced spreadability (O'Callaghan et al., 2016). The mean SI for CON samples was 1.76. The SI tended to be increased by PA starting at 150 g/d ($P = 0.05$) and was increased at 300 g/d and above ($P < 0.001$), with a maximal value of 2.24 at 750 g/d. Stearic acid decreased SI starting at

Table 1. The effect of feeding increasing doses of a fat supplement high in palmitic acid (PA) or stearic acid (SA) on milk fatty acid (FA) categories compared with a no-supplement control (CON)

	Treatment[1]				*P*-value[2]	
Milk FA category	CON	PA	SA	SEM	CON vs. PA	CON vs. SA
<16-C,[3] %						
150 g/d	28.5	27.4	28.2	0.519	0.008	0.38
300 g/d	28.8	26.7	27.7	0.519	<0.001	0.01
500 g/d	28.8	26.3	27.9	0.519	<0.001	0.02
750 g/d	28.8	25.0	27.4	0.529	<0.001	0.003
16-C,[4] %						
150 g/d	30.9	33.0	30.3	0.575	<0.001	0.12
300 g/d	31.2	34.7	29.7	0.575	<0.001	<0.001
500 g/d	31.3	36.5	29.0	0.575	<0.001	<0.001
750 g/d	31.3	38.6	28.0	0.584	<0.001	<0.001
>16-C,[5] %						
150 g/d	34.1	33.4	35.1	0.598	0.28	0.09
300 g/d	33.4	32.6	36.3	0.598	0.16	<0.001
500 g/d	33.2	31.4	37.2	0.598	0.002	<0.001
750 g/d	33.4	31.0	38.8	0.613	<0.001	<0.001
OBCFA,[6] %						
150 g/d	4.06	3.83	3.80	0.101	0.03	0.01
300 g/d	4.13	3.69	3.87	0.102	<0.001	0.01
500 g/d	4.21	3.41	3.65	0.104	<0.001	<0.001
750 g/d	4.10	3.24	3.43	0.104	<0.001	<0.001
SFA, %						
150 g/d	70.1	70.2	69.8	0.503	0.86	0.57
300 g/d	70.1	70.7	69.6	0.502	0.30	0.31
500 g/d	70.3	71.5	68.9	0.502	0.03	0.01
750 g/d	70.8	71.7	68.5	0.502	0.02	<0.001
UFA, %						
150 g/d	27.5	27.4	27.6	0.485	0.81	0.82
300 g/d	27.5	27.0	28.2	0.485	0.39	0.20
500 g/d	27.3	26.1	28.9	0.485	0.02	0.003
750 g/d	27.1	26.0	29.3	0.503	0.05	<0.001
Mean chain length, no.						
150 g/d	13.50	13.54	13.48	0.068	0.33	0.79
300 g/d	13.47	13.59	13.58	0.068	0.01	0.02
500 g/d	13.50	13.56	13.60	0.068	0.21	0.04
750 g/d	13.48	13.63	13.60	0.069	0.002	0.01

[1]Treatments were a no-supplement control (CON) or fat supplements high in palmitic acid (PA; 88% 16:0 and 8% 18:1*c*9) or stearic acid (SA; 81% 18:0 and 10% 18:1c9) fed at increasing doses every 4 d (150, 300, 500, or 750 g/d).

[2]Preplanned contrasts compared PA and SA to CON within each dose level.

[3]Sum of even, straight-chain FA <16-C.

[4]Sum of 16-C straight-chain FA.

[5]Sum of even, straight-chain FA >16-C.

[6]Odd- and branched-chain FA.

300 g/d (P < 0.001) to a minimum of 1.35 at the 750 g/d dose (Table 4).

Solid Fat Content at Key Temperatures

Solid fat content at 3°C, 20°C, and 38.6°C are specifically relevant to products stored at refrigerator and room temperature and milk fat during secretion in the mammary gland. For 20°C and 38.6°C, PA had increased SFC compared with CON starting at 300 g/d, whereas at 3°C, PA only increased SFC at 750 g/d. At 3°C and 20°C, SA decreased SFC compared with CON starting at 500 g/d, whereas no effect of SA was found at 38.6°C (Figure 3).

At refrigerator temperature (3°C), milk fat from the 750 g/d dose was 76.2% solid for CON, compared with 77.6% solid for PA (P < 0.05) and 73.9% for SA (P < 0.01). At room temperature (20°C), milk fat of the 750 g/d dose was 31.3% solid for CON, compared with 36.6% solid for PA (P < 0.05) and 28.1% for SA (P < 0.01). At cow body temperature (38.6°C), milk fat of the 750 g/d dose was 0.015% solid for CON, compared with 0.063% solid for PA (P < 0.001) and 0.008% for SA (P = 0.14).

Table 2. The effect of feeding increasing doses of a fat supplement high in palmitic acid (PA) or stearic acid (SA) on the profile of selected milk short- and medium-chain fatty acids compared with a no-supplement control (CON)

	Treatment[1]				P-value[2]	
FA, % FA	CON	PA	SA	SEM	CON vs. PA	CON vs. SA
4:0						
150 g/d	4.44	4.35	4.48	0.104	0.34	0.66
300 g/d	4.44	4.34	4.43	0.101	0.30	0.90
500 g/d	4.32	4.47	4.45	0.101	0.11	0.17
750 g/d	4.44	4.50	4.53	0.104	0.52	0.36
6:0						
150 g/d	2.48	2.41	2.50	0.054	0.27	0.66
300 g/d	2.51	2.37	2.45	0.054	0.01	0.27
500 g/d	2.46	2.37	2.46	0.054	0.10	0.91
750 g/d	2.49	2.30	2.51	0.056	<0.001	0.79
8:0						
150 g/d	1.40	1.33	1.42	0.037	0.05	0.62
300 g/d	1.43	1.30	1.4	0.037	<0.001	0.12
500 g/d	1.42	1.29	1.39	0.037	<0.001	0.49
750 g/d	1.42	1.21	1.40	0.037	<0.001	0.61
10:0						
150 g/d	3.34	3.10	3.29	0.107	0.004	0.58
300 g/d	3.39	3.01	3.20	0.107	<0.001	0.01
500 g/d	3.41	2.96	3.22	0.107	<0.001	0.02
750 g/d	3.38	2.70	3.16	0.108	<0.001	0.01
12:0						
150 g/d	3.91	3.61	3.79	0.127	0.002	0.206
300 g/d	3.99	3.50	3.72	0.127	<0.001	0.004
500 g/d	4.06	3.44	3.75	0.127	<0.001	0.001
750 g/d	3.98	3.10	3.58	0.129	<0.001	<0.001
14:0						
150 g/d	11.5	11.0	11.3	0.165	0.001	0.18
300 g/d	11.5	10.7	11.2	0.165	<0.001	0.02
500 g/d	11.6	10.4	11.2	0.165	<0.001	0.008
750 g/d	11.5	9.84	10.9	0.168	<0.001	<0.001
14:1*c*9						
150 g/d	1.06	1.04	1.01	0.038	0.80	0.36
300 g/d	1.09	1.02	1.01	0.038	0.13	0.08
500 g/d	1.12	1.00	1.00	0.038	0.01	0.01
750 g/d	1.06	0.97	0.94	0.040	0.03	0.01

[1]Treatments were a no-supplement control (CON) or fat supplements high in palmitic acid (PA; 88% 16:0 and 8% 18:1*c*9) or stearic acid (SA; 81% 18:0 and 10% 18:1c9) fed at increasing doses every 4 d (150, 300, 500, or 750 g/d).

[2]Preplanned contrasts compared PA and SA to CON within each dose level.

Melting Curve Parameters

Supplementing PA dose-dependently increased both peak melting temperature and end melting temperature of milk fat compared with CON starting at 300 g/d (P < 0.001; Figure 3). At 750 g/d, PA increased peak melting temperature 1.5°C and increased end melting temperature 1.8°C compared with CON. However, SA did not affect peak or end melting temperature at any dose. No effect of treatment was found at any dose on total enthalpy (data not shown).

Solid Fat Curves

Solid fat content was tested from 0°C to 40°C with data analyzed within dose with a model testing the effect of treatment, temperature, and their interaction (Figure 4). At the 150 and 300 g/d doses, we found an effect of treatment and temperature (both P < 0.001), but no interaction. The SFC of PA was consistently higher than CON at both doses (P = 0.02). The average SFC of SA was lower than CON at the 150 g/d dose (P = 0.03) but did not differ at the 300 g/d dose (P = 0.97).

At the 500 and 750 g/d doses, we observed an interaction between treatment and temperature (P < 0.001), where PA increased SFC of milk fat compared with CON from 15°C to 40°C (P < 0.001 for all), but not at lower temperatures. At the 500 g/d dose, SFC was lower (P < 0.05) in SA compared with CON at most temperatures 20°C and below, whereas at the 750 g/d dose, this expanded to include most temperatures between 0°C and 40°C.

Table 3. The effect of feeding increasing doses of fat supplements high in palmitic acid (PA) or stearic acid (SA) on the concentration of selected milk fatty acids (FA) compared with a no-supplement control (CON)

	Treatment[1]				*P*-value[2]	
FA, % of FA	CON	PA	SA	SEM	CON vs. PA	CON vs. SA
16:0						
150 g/d	29.5	31.5	29.0	0.544	<0.001	0.16
300 g/d	29.8	33.1	28.4	0.544	<0.001	0.001
500 g/d	29.8	34.9	27.8	0.544	<0.001	<0.001
750 g/d	29.9	36.8	26.9	0.552	<0.001	<0.001
16:1*c*9						
150 g/d	1.37	1.48	1.29	0.048	0.04	0.15
300 g/d	1.43	1.56	1.25	0.048	0.02	<0.001
500 g/d	1.45	1.61	1.20	0.048	0.003	<0.001
750 g/d	1.42	1.73	1.13	0.050	<0.001	<0.001
18:0						
150 g/d	9.44	9.07	10.1	0.317	0.29	0.06
300 g/d	8.96	8.60	10.3	0.317	0.30	<0.001
500 g/d	8.90	8.33	10.8	0.317	0.10	<0.001
750 g/d	9.27	8.06	11.9	0.328	<0.001	<0.001
18:1*c*9						
150 g/d	17.4	17.4	17.8	0.338	0.89	0.27
300 g/d	17.2	17.0	18.4	0.338	0.66	<0.001
500 g/d	17.2	16.6	19.3	0.338	0.08	<0.001
750 g/d	17.3	16.6	20.1	0.348	0.07	<0.001
18:2*c*9,*c*12						
150 g/d	1.96	1.91	1.95	0.092	0.25	0.69
300 g/d	1.94	1.87	1.91	0.093	0.18	0.61
500 g/d	1.91	1.78	1.93	0.092	0.01	0.54
750 g/d	1.89	1.74	1.96	0.093	0.004	0.16
18:3*c*9,*c*12,*c*15						
150 g/d	0.448	0.425	0.438	0.013	0.14	0.50
300 g/d	0.461	0.435	0.470	0.013	0.11	0.58
500 g/d	0.451	0.393	0.453	0.013	<0.001	0.94
750 g/d	0.416	0.365	0.440	0.013	0.002	0.15
CLA*c*9,*t*11						
150 g/d	0.380	0.384	0.374	0.020	0.65	0.80
300 g/d	0.370	0.390	0.370	0.020	0.44	0.28
500 g/d	0.368	0.358	0.356	0.019	0.49	0.98
750 g/d	0.356	0.365	0.323	0.019	0.74	0.28

[1]Treatments were a no-supplement control (CON) or fat supplements high in palmitic acid (PA; 88% 16:0 and 8% 18:1*c*9) or stearic acid (SA; 81% 18:0 and 10% 18:1c9) fed at increasing doses every 4 d (150, 300, 500, or 750 g/d).

[2]Preplanned contrasts compared PA and SA to CON within each dose level.

Linear Regression with Solid Fat Content of Milk Fat

Simple linear regression of SFC of milk fat at 20°C with individual FA identified 3 FA with an R^2 >0.40, including 16:0 (R^2 = 0.57), *i*17:0 (R^2 = 0.41), and 18:1*c*9 (R^2 = 0.44; $P < 0.001$ for all). Both 16:0 and 18:1*c*9 were >1% of milk FA and were analyzed further. Every 1-percentage-unit increase in milk fat 16:0 was associated with a 0.92-unit increase in the percentage solid at 20°C (solid at 20°C = 3.369 + 0.924 × 16:0; $P < 0.001$; Figure 5A). When the random effect of cow and period were added to the model, the full model adjusted R^2 increased from 0.57 to 0.86, but the partial R^2 of 16:0 was 0.61 (solid at 20°C = 6.016 + 0.840 × 16:0; $P < 0.001$; Figure 5C). Every 1-percentage-unit increase in 18:1c9 in milk FA was associated with a 1.39-percentage-unit decrease in SFC at 20°C (SFC at 20°C = 56.30 − 1.389 × 18:1c9; $P < 0.001$; Figure 5B) and the full model adjusted R^2 increased from 0.43 to 0.73 with the addition of the random effect of cow and period, although the partial R^2 of 18:1c9 was 0.48 (SFC at 20°C = 56.14 − 1.376 × 18:1c9; $P < 0.001$; Figure 5D).

Simple linear regression was also conducted with select FA indicators of interest to determine their relationship with predicted butter spreadability. We observed a relatively strong linear relationship between butter SI and SFC of milk fat at 20°C ($P < 0.001$, $R^2 = 0.61$; Figure 5E). However, we found no relationship between mean carbon chain length and SFC of milk fat at 20°C ($P > 0.05$; data not shown).

Table 4. The effect of feeding increasing doses of fat supplements high in palmitic acid (PA) or stearic acid (SA) on milk fat desaturase and spreadability indexes compared with a no-supplement control (CON)

Milk fat index	Treatment[1]				*P*-value[2]	
Dose	CON	PA	SA	SEM	CON vs. PA	CON vs. SA
16-C desaturase index[3]						
150 g/d	4.42	4.49	4.44	0.123	0.68	0.94
300 g/d	4.59	4.49	4.25	0.123	0.47	0.02
500 g/d	4.61	4.39	4.17	0.123	0.13	0.003
750 g/d	4.53	4.48	4.05	0.127	0.70	0.002
18-C desaturase index[4]						
150 g/d	64.6	65.7	63.9	0.832	0.25	0.50
300 g/d	65.4	66.3	64.1	0.832	0.34	0.16
500 g/d	65.6	66.4	64.3	0.832	0.39	0.16
750 g/d	64.7	67.3	63.0	0.862	0.009	0.08
Butter spreadability index[5]						
150 g/d	1.73	1.84	1.66	0.051	0.02	0.14
300 g/d	1.77	1.98	1.57	0.051	<0.001	<0.001
500 g/d	1.76	2.11	1.46	0.051	<0.001	<0.001
750 g/d	1.76	2.24	1.35	0.052	<0.001	<0.001

[1]Treatments were a no-supplement control (CON) or fat supplements high in palmitic acid (PA; 88% 16:0 and 8% 18:1*c*9) or stearic acid (SA; 81% 18:0 and 10% 18:1c9) fed at increasing doses every 4 d (150, 300, 500, or 750 g/d).

[2]Preplanned contrasts compared PA and SA to CON within each dose level.

[3]Defined as 16:1*c*9/(16:1*c*9 + 16:0).

[4]Defined as 18:1*c*9/(18:1*c*9 + 18:0).

[5]Defined as 16:0/18:1*c*9.

Backward Multivariate Regression

Of the 14 FA found at >1% of FA in milk fat, only 6:0, 18:0, 16:0, 18:1*c*9, 14:1*c*9, 10:0, 12:0, and 14:0 were retained in the final modeling predicting SFC at room temperature (all $P < 0.001$ and listed in order of significance; Table 5). The model had an R^2 of 0.962 and an adjusted R^2 of 0.959. Backward regression was also conducted with the same 14 FA as a molar percentage of FA. The FA that remained significant in the model included 18:0, 6:0, 16:0, 18:1*c*9, 14:1*c*9, 10:0, 12:0, and 14:0 (all $P < 0.001$; listed in order of significance; data not shown). The model had an R^2 of 0.961 and an adjusted R^2 of 0.959.

Principal Component Analysis

The first 4 principal components (**PC**) explained 26.7%, 18.3%, 13.8%, and 7.6% of the variation (total 66.5%). Overall, there were 9 PC with eigenvalues greater than 1 that explained 87.0% of the variance. The most influential component in PC1 was 18:1*t*9, whereas the most influential component in PC2 was *i*15:0 (Figure 6).

The variable most closely related to SFC at 20°C within PC1 and PC2 was 16:0 concentration. This relationship continued in PC3 and PC4, where in all combinations of 2-way analyses between PC1 through PC4 using the Loading Plot function in JMP (PC1 vs. PC2, PC1 vs. PC3, PC1 vs. PC4, PC2 vs. PC3, PC2 vs. PC4, and PC3 vs. PC4), SFC at 20°C was most closely related to the 16:0 percentageof milk FA (data not shown). The desaturated product of 16:0, which is 16:1*c*9, was also in the same quadrant as SFC at 20°C for all combinations of 2-way analyses between PC1 through PC4. The other group of FA that were in the same quadrant as SFC at 20°C for the primary analysis of PC1 versus PC2 was the medium chain OBCFA (9:0, 11:0, 13:0, and 15:0), which all clustered together between 16:0 and 16:1*c*9 and at approximately the same vector length. This relationship did not remain consistent for comparing PC1 and PC2 with PC3 and PC4.

The variable most directly opposite to SFC at 20°C in the plot of PC1 versus PC2 was 18:1*t*11. This FA continued to remain opposite of SFC at 20°C in all combinations of 2-way analyses between PC1 through PC4. Additionally, 18:0 was in the opposite quadrant or on the border close to being in the opposite quadrant from SFC at 20°C for PC1 through PC4, although the vector length was smaller. Lastly, 18:1*c*9 was between ~120° to 180° around the circle from SFC at 20°C for all combinations of PC1 through PC3, but this relationship did not hold when comparing these principal components to PC4.

DISCUSSION

Strategies of fat feeding to dairy cows are largely based on increasing energy intake and milk and milk fat yield, with little attention given to the impacts on milk fat characteristics and functional properties important to

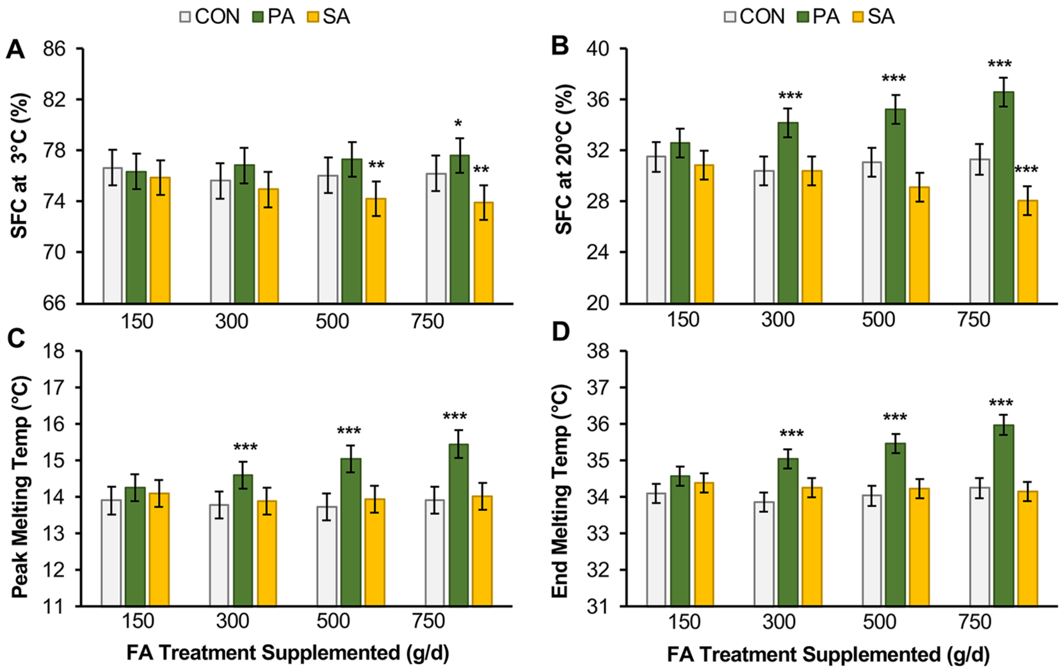

Figure 3. The effect of feeding fat supplements high in palmitic acid (PA) or stearic acid (SA) fed at increasing doses every 4 d (150, 300, 500, or 750 g/d) on solid fat content (SFC) at 3°C (A) or 20°C (B) as well as peak melting temperature (C) and end melting temperature (D) of milk fat compared with a no-supplement control (CON). Solid fat content at each temperature was calculated for each treatment as (total enthalpy – enthalpy up to X°C)/total enthalpy. Data are presented as LSM with SEM bars. Asterisks indicate treatment differences from control within each dose (*$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$). Temp = temperature.

dairy products. Controlled experiments commonly feed dry fat supplements at 1.5% to 2% of the diet (375 to 500 g/d at 25 kg DMI), which is considered the upper range of practical feeding levels in the United States. However, feeding levels of the supplements studied vary considerably in the field, and few dose titrations are reported in the literature (Boerman et al., 2017). The upper limit to FA supplementation is affected by disruption of rumen fermentation when feeding UFA and limitations in FA absorption, and NASEM (2021) recommends total dietary FA to be ≤7% of diet DM. The 750 g/d dose in the current trial was 2.8% of diet DM and was selected to allow characterization above current field feeding levels. Each dose was fed for 4 d. However, with the dose escalation design, as the dose increased, so did the total time exposed to the supplement. The time required for SFA supplements to change milk FA profile has not been well described, but is expected to be short because SFA are not modified by the rumen, absorbed FA are cleared rapidly from the blood, and palmitic acid does not affect mammary expression of lipogenic genes (Bernard et al., 2021; Haile et al., 2024). Our main objective was to provide milk fat with a wide range of palmitic acid concentrations, which was accomplished, as discussed further in this section, and demonstrates that that the 4-d period was adequate to result in changes in milk FA. However, care should be taken to extrapolate that long-term feeding of specific doses may have a larger effect if the 4-d period did not result in stabilization of the milk FA profile.

Palmitic acid supplements increase palmitic acid in milk fat, as it is not modified in the rumen and elongation and desaturation are limited (Loften et al., 2014). The meta-analysis on 16:0 feeding conducted by dos Santos Neto et al. (2021) found that every 1-percentage-unit increase in feeding high palmitic supplements increased milk 16-C FA concentration by 3.6 percentage units. In the current experiment, PA similarly dose-dependently increased milk 16-C FA, and the 300 g/d dose that increased dietary PA ~1.1 percentage units increased milk 16-C FA by 3.5 percentage units compared with CON.

Fewer experiments have investigated the effects of high stearic acid supplements, as most commercial products are high in palmitic acid or contain a blend of palmitic and stearic acid. Additionally, many of the experiments testing high stearic acid supplements used highly pure stocks almost devoid of UFA and have very low digestibility (e.g., Boerman et al., 2017). Shepardson and Harvatine (2021)fed a 93% stearic and 5% oleic acid supplement more similar to the current study and observed no change in concentration of <16-C or 16-C FA, but the supplement increased both stearic acid

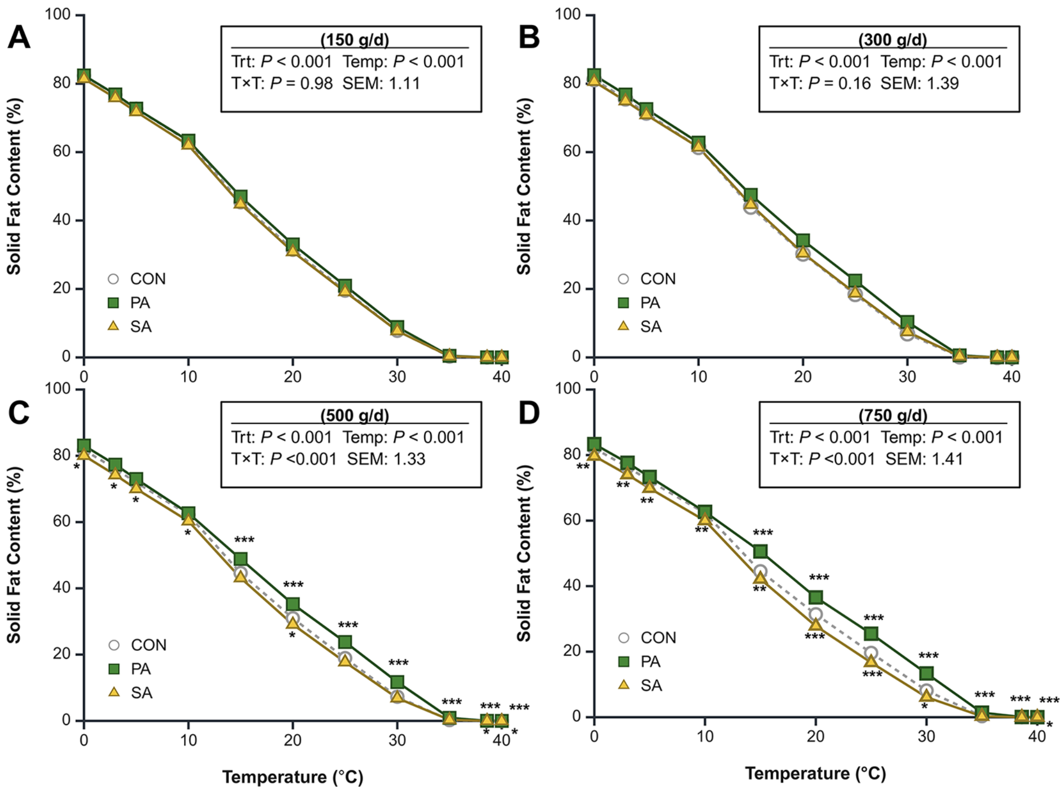

Figure 4. The effect of feeding fat supplements high in palmitic acid (PA) or stearic acid (SA) fed at increasing doses every 4 d (150 g/d [A], 300 g/d [B], 500 g/d [C], or 750 g/d [D]) on solid fat content of milk fat across different temperatures compared with a no-supplement control (CON). Solid fat content at each temperature was calculated for each treatment as (total enthalpy – enthalpy up to X°C)/total enthalpy. Data are presented as LSM. Asterisks above the line indicate differences between PA and CON, and asterisks below indicate differences between SA and CON at each temperature (*$P < 0.05$, **$P < 0.01$, and ***$P < 0.001$). Trt = treatment; temp = temperature. Created with BioRender.com. Homan, A. (2025); https://biorender.com/b8mf7hb.

and oleic acid by 1.6 percentage units. The decrease in the concentration of <16 C and 16-C FA at the 500 g/d dose in the current experiment and the 4-percentage-unit increase of >16-C FA may indicate that the higher level of oleic acid in the SA supplement improved FA digestibility, although this was not measured. The melting temperatures of palmitic and stearic acid are 63°C and 70°C, whereas oleic acid is 16°C, and the changes in FA profile observed are consistent with the directional changes in melting properties observed.

Stearoyl-CoA desaturase (**SCD**) catalyzes the conversion palmitic and stearic acids to monounsaturated palmitoleic and oleic acid, respectively, and the enzyme has a higher activity for stearic acid as its substrate compared with palmitic acid (Ntambi and Miyazaki, 2004). Mosley and McGuire (2007) provided the most direct quantification of FA desaturation using boluses and continuous abomasal infusions of ^{13}C-labeled palmitic and stearic acid and reported that 49% to 60% of stearic acid and 2.5% to 2.6% of palmitic acid were desaturated. Similarly, Toral et al. (2017) reported 47% to 57% conversion of i.v.-infused ^{13}C-labeled stearic acid to oleic acid in lactating ewes. Enjalbert et al. (1998) calculated mammary desaturase activity using arteriovenous difference and milk concentration of stearic and oleic acid and found that ~52% of stearic acid is desaturated in the mammary gland. As a result of the higher efficiency of SCD for stearic acid, increasing dietary stearic acid will lead to substantial increases in both stearic and oleic acid, whereas feeding palmitic acid will primarily increase palmitic acid, as observed in our current experiment.

The desaturase index is an approximation of SCD activity in the mammary gland (Kelsey et al., 2003), as it is the ratio of products to precursors for the enzyme. The average desaturase index for 16-C FA across CON samples in the current study was similar to what has been reported in the literature, whereas the average desaturase index for 18-C FA was ~2.5 to 4 percentage units lower than other studies (Schennink et al., 2008; Kgwatalala et al., 2009) and may be due to differences in parity or genetics. Increasing SA dose in the current experiment decreased both 16-C and 18-C desaturase indices, which is expected based on enzyme kinetics. The lack of response of the 16-C desaturase index with

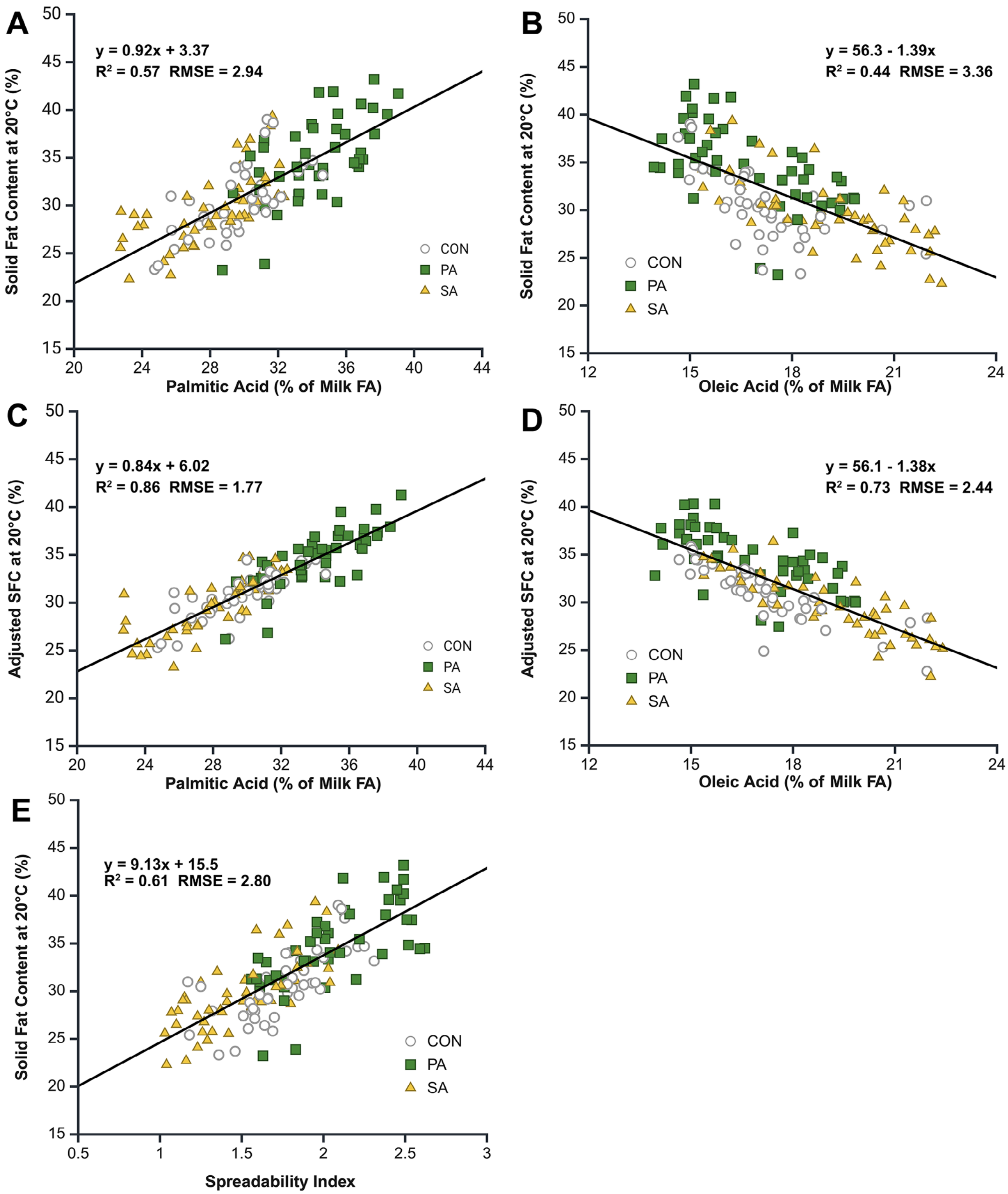

Figure 5. The relationship between palmitic or oleic acid concentration in milk fat and solid fat content (SFC) of milk fat at 20°C. Panel A and panel B show simple linear regressions of palmitic and oleic acid with SFC at 20°C, respectively. Panel C and panel D show random regression of SFC at 20°C versus palmitic (C) and oleic (D) acid while also modeling the random effect of cow and period (plotted data adjusted for the random effects). Panel E shows the relationship between spreadability index, defined as the ratio of palmitic to oleic acid (16:0/18:1*c*9) with SFC at 20°C. All regression analyses had a *P*-value <0.001. Treatments were fat supplements high in palmitic acid (PA) or stearic acid (SA) fed at increasing doses every 4 d (150, 300, 500, and 750 g/d) and a no-supplement control (CON). RMSE = root mean square error. Created with BioRender.com. Homan, A. (2025); https://biorender.com/9pfk5yh.

Table 5. Backward regression analysis of the effect of the concentration of individual milk fatty acids[1] on solid fat content of milk fat at 20°C

Variable[2]	Estimate	SE (Est)	*P*-value	Partial R^2
Intercept	51.37	6.03	<0.001	
18:0	2.18	0.10	<0.0001	0.1404
6:0	−11.86	0.55	<0.0001	0.1395
16:0	0.98	0.07	<0.0001	0.0648
18:1*c*9	−1.67	0.12	<0.0001	0.0580
14:1*c*9	5.44	0.61	<0.0001	0.0246
10:0	11.93	1.41	<0.0001	0.0219
12:0	−7.29	1.21	<0.0001	0.0110
14:0	−0.70	0.19	0.0003	0.0043

[1]The initial model included all milk fatty acids with concentrations greater than 1% (14 total). The model was reduced by removing the least significant fatty acid one at a time until all model factors had a *P*-value of ≤0.01.

[2]Fatty acids are listed in order of significance in the model.

PA is not clear and is in opposition to observations made by Rico et al., (2014), who reported an increase in the 16-C desaturase index and no change in 18-C index with palmitic acid supplementation. The SCD enzyme is highly responsive to changes in nutrition and metabolic hormones, and PA may increase mammary expression or activity of SCD under certain dietary conditions to partially compensate for increased milk saturated FA. Palmitic and stearic acid both increased SCD expression in preadipocyte cells isolated from steers (Choi et al., 2015), but had no effect in a bovine mammary cell line (MAC-T; Jacobs et al., 2013).

In the current experiment, milk FA profile was analyzed by solvent extraction of whole milk, and the melting temperature analysis was conducted on milk fat extracted by centrifugation. Milk FA profile was analyzed before initiating work to characterize melting properties, and the FA profile of the milk fat was not directly measured because only minor differences in FA profile have been reported between milk and butter FA profile (Baer et al., 2001; Bobe et al., 2003). Melting analysis was also conducted on milk fat extracted by centrifugation rather than butter, as it was not practical to make butter from each cow at each dose level due to the time and cost associated with doing so. Additionally, extracted milk fat provides a convenient standard sample to understand the influence of changes in FA profile on milk fat melting properties without confounding factors during butter processing that could potentially influence melting temperature. Furthermore, Chrusciel et al. (2023) reported a very strong correlation between butter spreadability and melting characteristics of butterfat measured by DSC (r >0.96), supporting the use of DSC to characterize physical properties of butter. It is important to recognize that some postharvest aspects of manufacturing also influence physical properties of butter, including temperatures during churning and temperature and length of storage (Huebner and Thomsen, 1957). It is possible that there are interactions between milk FA profile and processing methods, where postharvest factors could potentially compensate for changes in milk FA profile. Nelson et al. (2025) reported that although both higher de novo FA and UFA decrease milk fat melting temperature, milk fat with higher de novo FA was more responsive to softening by temperature cycling than milk fat with higher UFA. However, Landry et al., (2024) did not observe an interaction between palmitic acid supplementation and cream ripening procedures, concluding that the cold-warm-cold cream ripening procedure could not completely compensate for increased butter firmness caused by dietary palmitic acid supplementation. Further research will be needed to determine other interactions, including the impact of FA position on the glycerol backbone.

Milk fat properties have been well studied in dairy processing research, and the overall melting patterns of milk fat in this experiment were similar to those of Tomaszewska-Gras (2013) under the same method. However, there has been limited work on the effect of dairy nutrition in general, and specifically palmitic and stearic supplements, on milk fat melting properties (Harvatine, 2021). Enjalbert et al. (2000) observed that postruminal infusion of ~500 g/d of a high stearic acid supplement tended to decrease SFC of milk fat at 30°C and below, based on DSC analysis. This was similar to the current experiment, where 500 g/d of SA was also more liquid than CON at 20°C and below. Although stearic acid has a high melting temperature, the decrease in milk fat melting temperature is likely due to the near-equal increase in stearic and oleic acid with these treatments. We are not aware of other work investigating the direct effect of stearic acid on melting properties, but both Enjalbert et al. (2000) and Ortiz Gonzalez et al. (2024) report that postruminal infusions of 500 g/d of a high oleic acid oil drastically decreased SFC to approximately a third of the SFC of control samples at temperatures ~20°C.

The effect of palmitic acid on melting properties is more well described (summarized by Harvatine, 2021). Enjalbert et al. (2000) reported that postruminal infusion of ~500 g/d of a high palmitic acid supplement increased SFC of milk fat using DSC starting at 18°C compared with a no fat infusion control. However, the magnitude of the response at 18°C was over twice that observed at 20°C in the current work, likely due to a greater increase in milk palmitic acid concentration (46.2% vs. 34.9%, respectively). The difference may be due to differences in basal diet FA concentrations or digestibility of the FA supplements. Ortiz-Gonzalez et al. (2007) also reported an 11-percentage-unit increase in SFC of milk fat by DSC analysis with infusion of ~450 g/d of high palmitic acid soy oil (~40% 16:0) compared with conventional soy oil

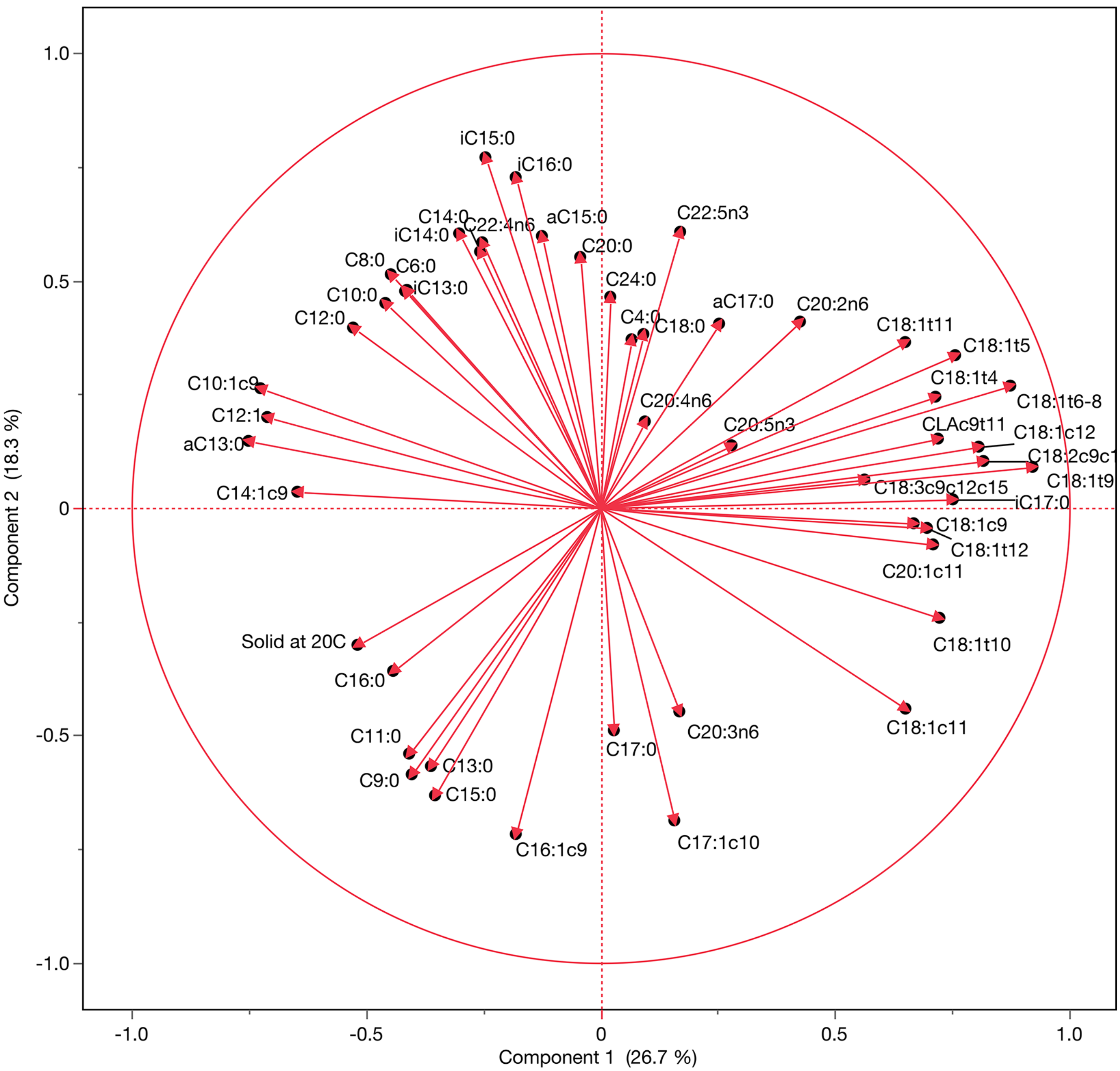

Figure 6. Principal component analysis of the relationship between the concentration of individual fatty acids in milk fat and solid fat content (SFC) of milk fat at 20°C from cows fed fat supplements high in palmitic acid (PA) or stearic acid (SA) at increasing doses every 4 d (150, 300, 500, and 750 g/d) or a no-supplement control (CON). Total n = 135.

(~9% 16:0), but milk fat enrichment of palmitic acid was lower than the current work, as expected.

Others have reported similar findings with feeding palmitic acid supplements using alternative measurements of butter hardness, but these cannot be directly compared with the current experiment due to the different analytical techniques used. Similar to the current experiment, (Relling et al., 2022b) reported that increasing levels of a high palmitic acid supplement linearly increased milk fat hardness. Chamberlain and DePeters (2017) also reported that a high palmitic acid (79%) supplement fed at 2% of diet DM increased butter hardness at room temperature and increased peak melt by 2°C compared with a yellow grease supplement low in palmitic acid. Bobe et al. (2003) also observed that butter samples with a higher atherogenic index that included higher palmitic acid concentrations were harder and more adhesive at both 5°C and 23°C than those with a lower atherogenic index. Interestingly, Landry et al., (2025) observed a large increase in butter hardness (+69%) at 20°C with a high palmitic acid supplement fed at 2% of diet DM, but did not observe a difference in SFC using DSC until 30°C.

It is unclear whether the very small increase in milk SFC at cow body temperature has biological relevance. It has been proposed that milk FA and triglyceride compositions are regulated by the mammary gland to maintain fluidity at body temperature (Loften et al., 2014), but the biological mechanism sensing and regulating this is not described. It is interesting to note that only trace amounts of solid fat were observed even at the highest doses of PA in the current experiment.

Regression and correlation analyses can also provide insight into the impact of FA profile on butter properties. Marangoni et al. (2022) analyzed the relationship between palmitic acid and butter hardness at 20°C in 40 different butter samples sourced from grocery stores across Canada. They reported a positive linear relationship (r = 0.74 and R^2 calculated to be 0.55) between palmitic acid concentration and mechanical firmness measured by the force required to penetrate butter. The samples in their dataset ranged from 31.9% to 39.8% palmitic acid. The dose titration treatments in the current study provided a broader range of 16:0 from 22.7% to 39.1% and regression analysis observed a similar positive linear relationship between palmitic acid and SFC of milk fat at 20°C by DSC (P < 0.001; R^2 = 0.57). Staniewski et al. (2021) also collected butter samples across Poland in both summer and winter months and determined that 16:0 had the highest negative correlation at 20°C (−0.70) with penetration depth, indicating harder butter, whereas oleic acid had a strong positive correlation of 0.84, indicating softer butter. Stearic acid also had a positive correlation similar to oleic acid (0.85) because of its strong positive correlation to milk unsaturated FA, especially oleic acid. This agrees with Enjalbert et al. (2000), who reported a strong negative relationship between oleic acid concentration and SFC of milk fat at 18°C based on DSC (R^2 = 0.93). A negative relationship between oleic acid and SFC at 20°C was also observed in the current experiment, but the R^2 was only 0.44. The stronger relationship in Enjalbert et al. (2000) was due to postruminal infusions of oleic acid, which greatly increased the range of oleic acid in their dataset compared with the current study (~15% to 40% vs. 13.9% to 22.4% of FA). Ortiz Gonzalez et al. (2024) also observed a linear decrease in SFC of milk fat at 20°C when increasing postruminal infusions of high oleic sunflower oil; oleic acid percent of milk fat ranged from 19% for the control to 57% at the highest dose of 1,000 g/d, and SFC decreased from 29.3% to 0.2% (FA profile reported in Drackley et al., 2007).

The SI has been proposed as a method to predict changes in butter softness based on milk FA profile and aligned directionally with the changes in melting temperature observed by DSC. This SI appears to be a simple indicator of SFC at room temperature, as it contains the only 2 FA with a concentration >1% of milk FA that had moderate individual relationships (R^2 >0.40) with SFC at room temperature, and it correlates well with SFC at room temperature in a direct regression analysis (R^2 = 0.61). However, a lower index number (a lower 16:0/18:1 ratio) is not always better; Staniewski et al. (2021) also reported that butter with desirable rheological properties was difficult to obtain if the SI was below 1.12. The experiments described previously with oleic acid infusions indicate an obvious potential to reduce SFC at room temperature and improve spreadability through increasing the oleic acid content of milk, but authors also caution that there are negative effects on butter functionality if its concentration is too high. Ortiz Gonzalez et al. (2024) concluded that postruminal infusions of 250 g/d were the limit to maintain butter functional properties, and that palmitic acid needed to be at least 20% of milk fat to maintain these properties. The current experiment provides additional insight into the potential of stearic acid to increase oleic acid in milk fat and decrease SFC at 20°C while also maintaining adequate saturated fat to retain expected physical properties. Feeding blends of palmitic, stearic, and oleic acid may also help to reduce the effects of palmitic acid on butter hardness if palmitic acid supplementation is desired. Relling et al. (2022b) observed that a high palmitic acid supplement increased butter hardness, whereas a supplement based on palm FA distillate that contained both palmitic and oleic acid decreased butter hardness. Additionally, factors that increase de novo FA synthesis in the mammary gland, such as acetate supplementation, increasing fiber digestibility, season, and genetics may also increase palmitic acid and could interact with supplementation strategies to alter melting properties.

Backward regression analysis of individual FA >1% of milk FA versus SFC at 20°C was conducted to understand if a combination of FA is a better predictor of solid butter than individual FA alone. Palmitic, stearic, and oleic acid were all highly significant contributors to the model, as expected. However, several of the <16-C FA, including 6:0, 10:0, 12:0, and 14:0, were also highly significant in the model, many of which had larger estimates than palmitic, stearic, and oleic acid. This was interesting given that none of these FA individually had an R^2 >0.07 in simple regression with SFC at 20°C. Furthermore, although 6:0, 12:0, and 14:0 had negative estimates, 10:0 had a positive estimate, implying that

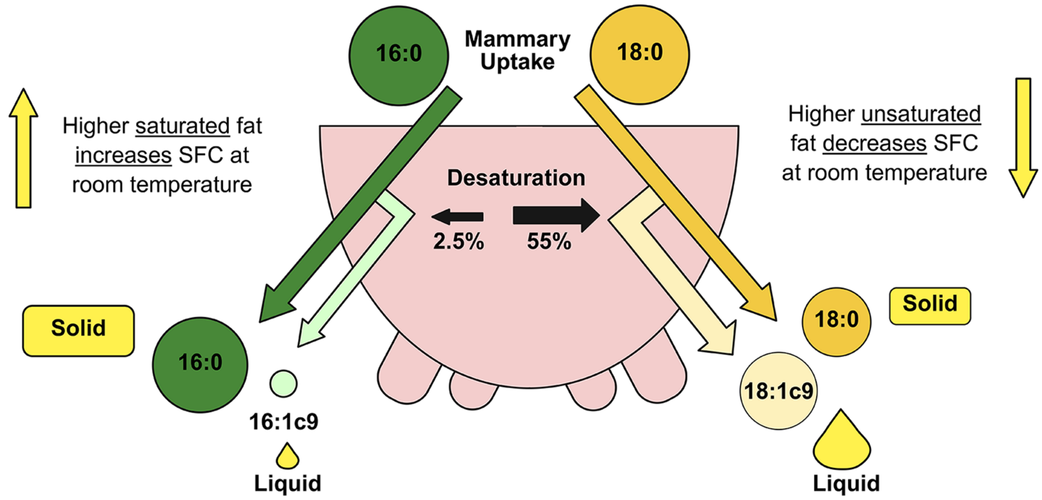

Figure 7. Summary of the effects of palmitic and stearic acid on mammary desaturase activity and solid fat content (SFC) of milk fat at room temperature. About 55% of stearic acid (18:0) is converted to oleic acid (18:1) in the mammary gland, whereas only about ~2.5% of palmitic acid (16:0) is converted to palmitoleic acid (16:1). This results in higher saturated fat content of milk when feeding palmitic acid and therefore an increase in SFC at room temperature, whereas feeding stearic acid increases unsaturated fat content of milk and decreases SFC at room temperature. Created with BioRender.com. Homan, A. (2025); https://biorender.com/15iqmyr.

they were not all working together in the model. It is unclear why this might be the case, given that increases in milk palmitic acid were associated with decreases for all FA <16-C except for 4:0 in the current experiment, but previous work has pointed out that palmitic acid supplements do not affect short- and medium-chain FA equally (Rico et al., 2014), and the FA position within the triglyceride may also have an effect. Backward regression analysis using molar concentration was also conducted because although most FA concentration analyses are done using g/100 g FA, shorter-chain FA may have a more profound effect on melting properties because they still occupy one-third of the positions for FA in a triglyceride. The results when analyzed by molar concentration were very similar to analysis by weight, likely because molar concentration is used to calculate FA concentration on a g/100 g basis.

Principal component analysis was conducted to further explore the relationship between the 48 identified individual FA in this set of milk samples and the SFC of milk fat at room temperature (20°C). Palmitic acid was the variable most closely associated with SFC at 20°C across all 2-way analyses between PC1 through PC4, which aligns with the linear relationship we observed between 16:0 and SFC at 20°C. Palmitoleic acid was also in the same quadrant, which is logical given that most milk palmitoleic acid comes from the small amount of desaturation of palmitic acid by SCD, as described earlier. The other FA in the same quadrant were many of the odd-chain FA, including 9:0, 11:0, 13:0, and 15:0. The reason for this is not clear, but it is possible that these odd-chain FA affect the crystallization of FA in milk fat in unique ways that would promote higher melting temperatures, or that there are correlations between odd-chain FA and other FA that influence melting temperature.

The FA more directly across from SFC at 20°C in the PC analysis were *trans* FA, as well as PUFA. These FA become diluted as palmitic acid increases in milk fat, which would explain their negative relationship. Polyunsaturated FA also have lower melting temperatures than MUFA. Surprisingly, none of the PUFA individually had a significant linear relationship with SFC at 20°C by bivariate analyses, which might be due to their lower abundance in milk fat. Branched-chain FA were also nearly opposite to SFC at 20°C in the PCA, likely due to their unique structures that would decrease the ability for FA to form tighter structures and ultimately affect milk fat crystallization.

Finally, although most even-chain <16-C FA (6:0, 8:0, 10:0, 12:0, and 14:0) were in an adjacent quadrant to SFC at 20°C, 4:0 was in an opposite quadrant. Butyrate is the short-chain FA in milk with the lowest melting temperature due to the fewest number of C–C bonds, which would help to counteract the effect of increasing palmitic acid on melting temperature. However, the 4:0 concentration in milk fat was not influenced by palmitic or stearic acid supplementation. Dietary strategies that increase milk 4:0 could be further explored as a potential to counteract increases in SFC with palmitic acid.

CONCLUSIONS

Fat supplementation altered milk FA profile and melting properties in a dose-dependent manner with no apparent plateau below 750 g/d. Palmitic acid supplementation increased milk palmitic acid concentration and SFC of milk fat at room temperature, whereas stearic acid supplementation increased milk stearic and oleic acid and decreased the SFC of milk fat at room temperature, likely due to large differences in mammary desaturase activity (summarized inFigure 7). Milk palmitic acid and oleic acid concentrations had the strongest relationship with SFC of milk fat at room temperature and can serve as predictors of butter melting properties. Further research with consumer preference panels with butter made from cows given different doses or combinations of these FA is needed to determine the point where increase in SFC of butter is noticeable. Understanding the effects of dietary fat and milk FA on butter melting properties can provide dairy producers and butter processors with strategies to better meet consumer preferences and increase product demands.

NOTES

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Nonstandard abbreviations used: CON = control; DSC = differential scanning calorimetry; FA = fatty acid; OBCFA = odd- and branched-chain FA; PA = supplement high in palmitic acid; PC = principal component; PCA = principal component analysis; RMSE = root mean square error; SA = supplement high in stearic acid; SCD = stearoyl-CoA desaturase; SFC = solid fat content; SI = spreadability index; temp = temperature; trt = treatment.

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